PROJECT REPORT PRESENTATION

Hate Speech Detection Using Machine Learning

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The Problem

Information Overload

The exponential growth of Web 2.0 has made manual moderation of hate speech computationally infeasible due to the sheer volume of data.

Core Challenges

- **Linguistic Ambiguity:** Sarcasm, slang, and context make detection difficult.
- **Data Imbalance:** Hate speech is often the minority class (16%), biasing models towards "Not Hate".

Dataset & Methodology

The Source

A large-scale dataset of approximately **842,000 comments** collected from Wikipedia Talk pages, offering a diverse range of conversational text.

Imbalance

The data reflects real-world skew: **84.1% Not Hate** vs. **15.9% Hate Speech**. This 5.3:1 ratio requires careful handling during training.

Preprocessing

Rigorous cleaning (URL removal, lowercasing) followed by **TF-IDF Vectorization** with **Bigrams** to capture contextual nuances.

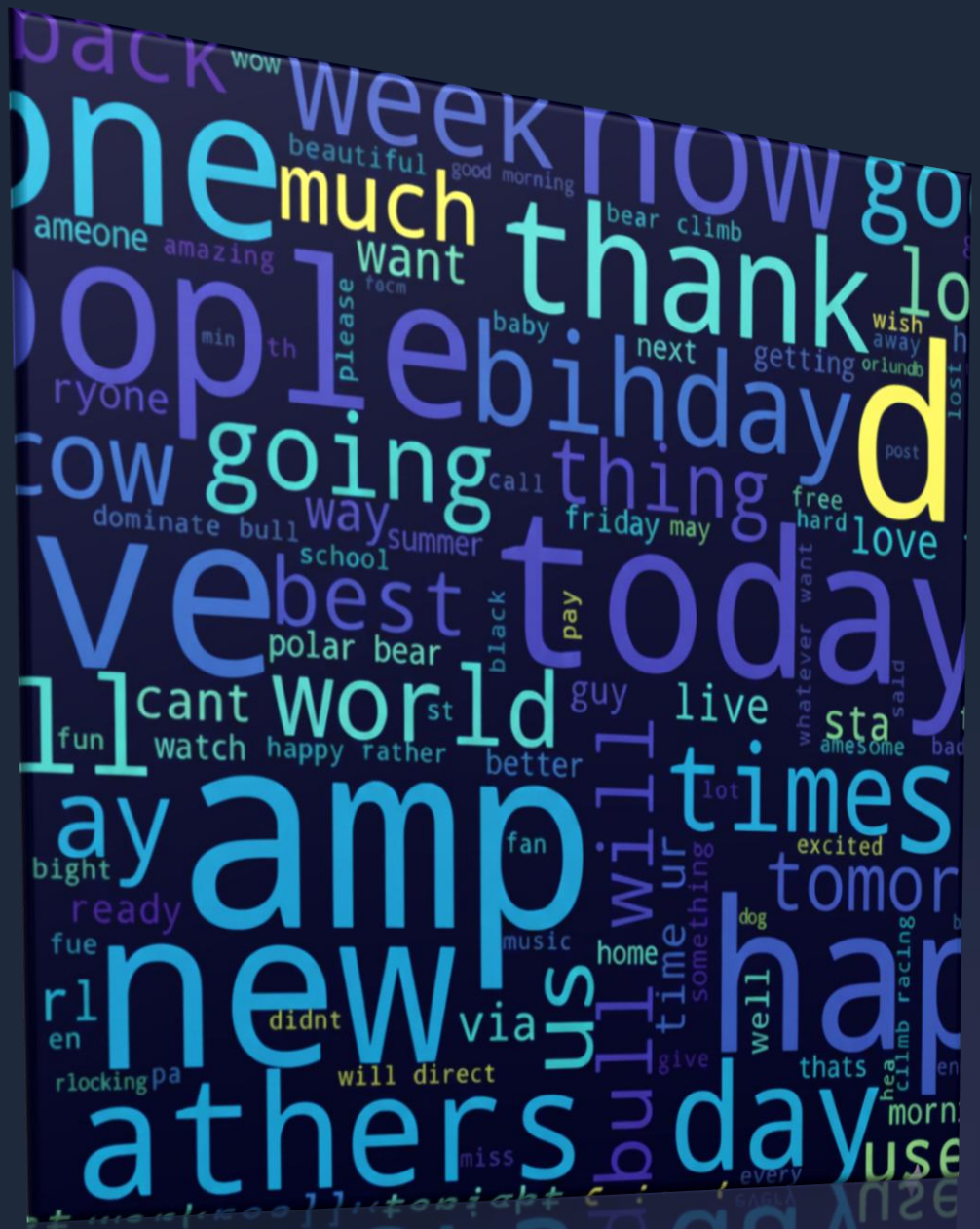
Word Visualization

Distinct Vocabularies

Not Hate: Dominated by Wikipedia-specific administrative terms like "article," "page," "edit," and "consensus."

Hate Speech: Characterized by explicit insults, racial slurs, and aggressive terms.

This clear lexical distinction suggests TF-IDF will be an effective feature extraction method.



Machine Learning Models

Multinomial Naive Bayes

Used as a baseline. It is extremely fast but assumes feature independence, leading to lower recall for complex hate speech.

Logistic Regression

A probabilistic linear classifier. It offers a balance between speed and accuracy but struggled slightly with the high class imbalance.

Linear SVM

The Winner. Creates an optimal hyperplane with maximum margin. Best suited for high-dimensional sparse data like text.

Model 1: Multinomial Naive Bayes

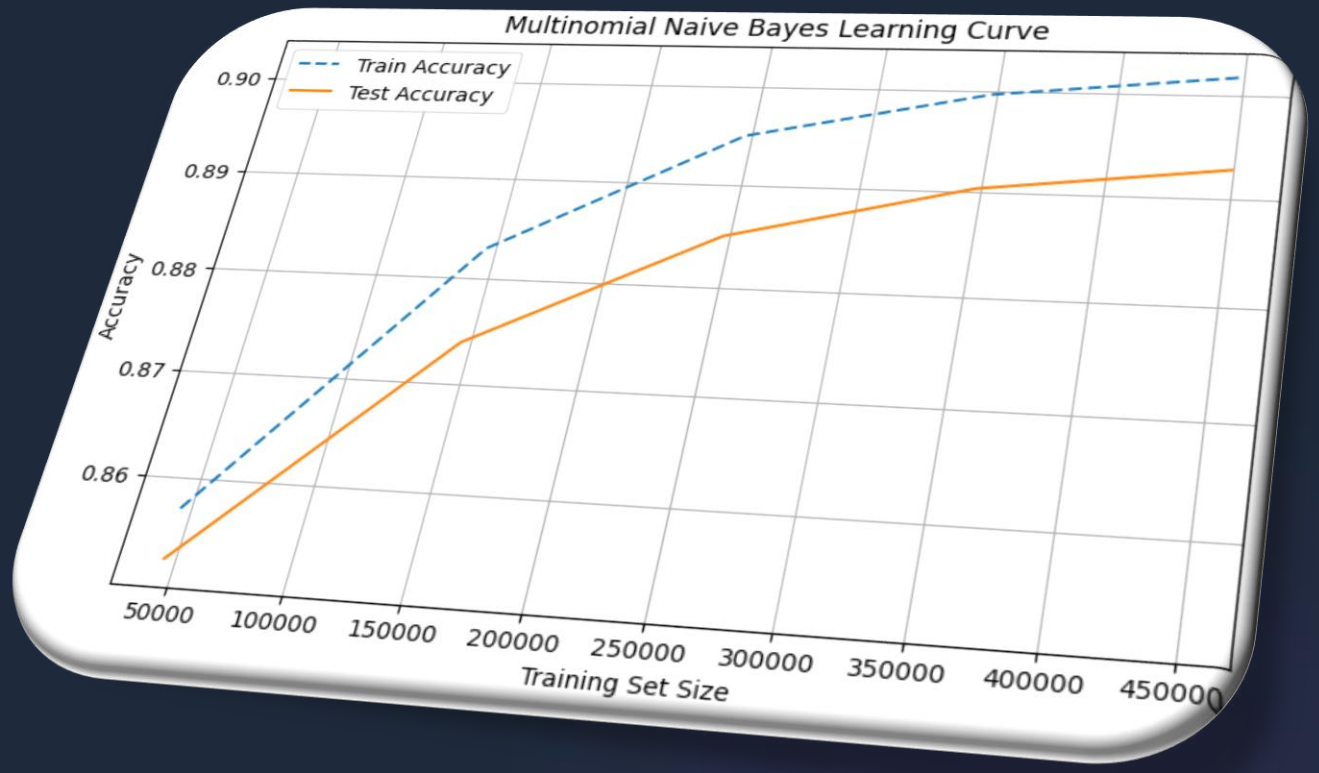
Class	Precision	Recall	F1-Score	Support
0 (Not Hate)	0.92	0.96	0.94	141,728
1 (Hate)	0.71	0.57	0.63	26,739
Accuracy	**0.89**			

Insight: High precision but very poor Recall (0.57) for the Hate class. It missed 43% of toxic comments.

Learning Curve

Training Analysis

- Extremely fast convergence due to probabilistic nature.
- Shows a gap between training and validation accuracy.
- Indicates potential bias/underfitting for the complex minority class.



Model 2: Logistic Regression

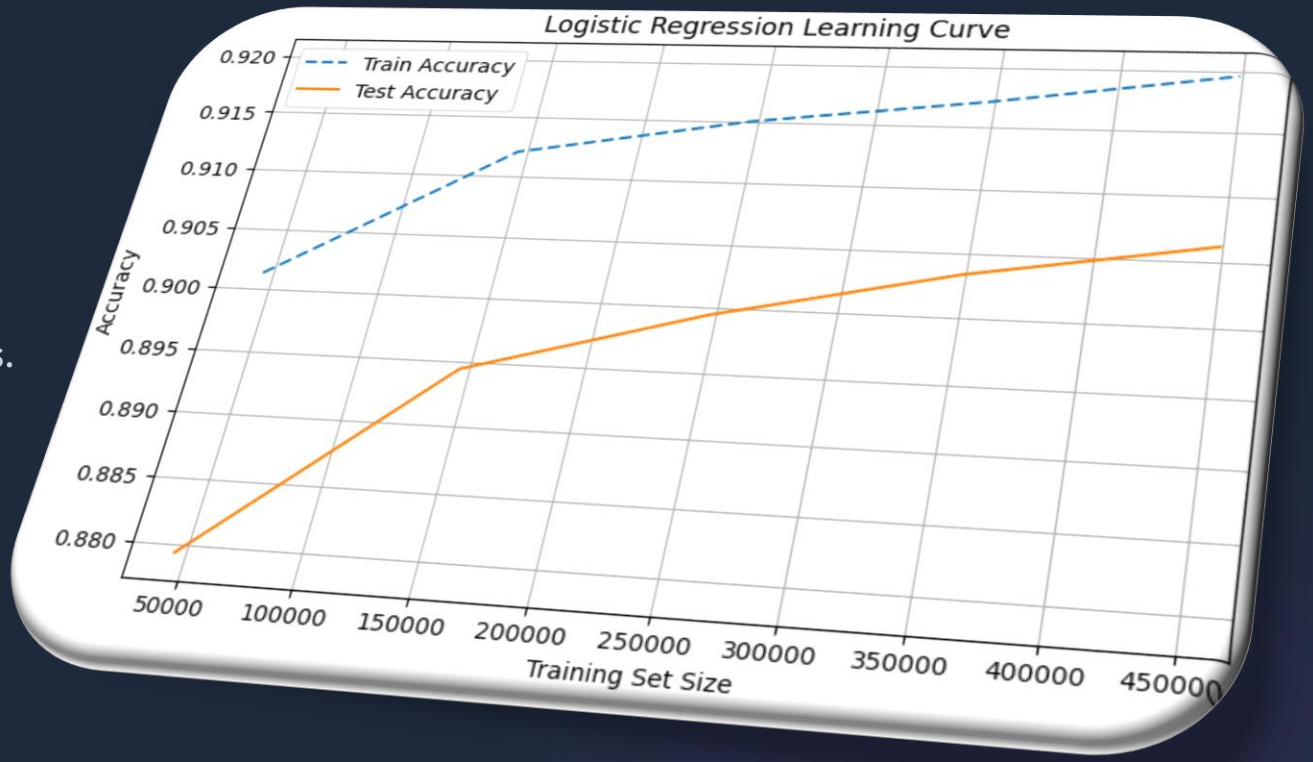
Class	Precision	Recall	F1-Score	Support
0 (Not Hate)	0.92	0.97	0.95	141,728
1 (Hate)	0.81	0.57	0.67	26,739
Accuracy	**0.90**			

Insight: Precision improved significantly to 0.81, but Recall remained low (0.57). It still misses too much toxicity.

Learning Curve

Training Analysis

- Smoother convergence compared to Naive Bayes.
- Required more iterations (2000) to reach optimal weights.
- Shows improved generalization but limited by the linear decision boundary on imbalanced data.



Model 3: Linear SVM (Best Model)

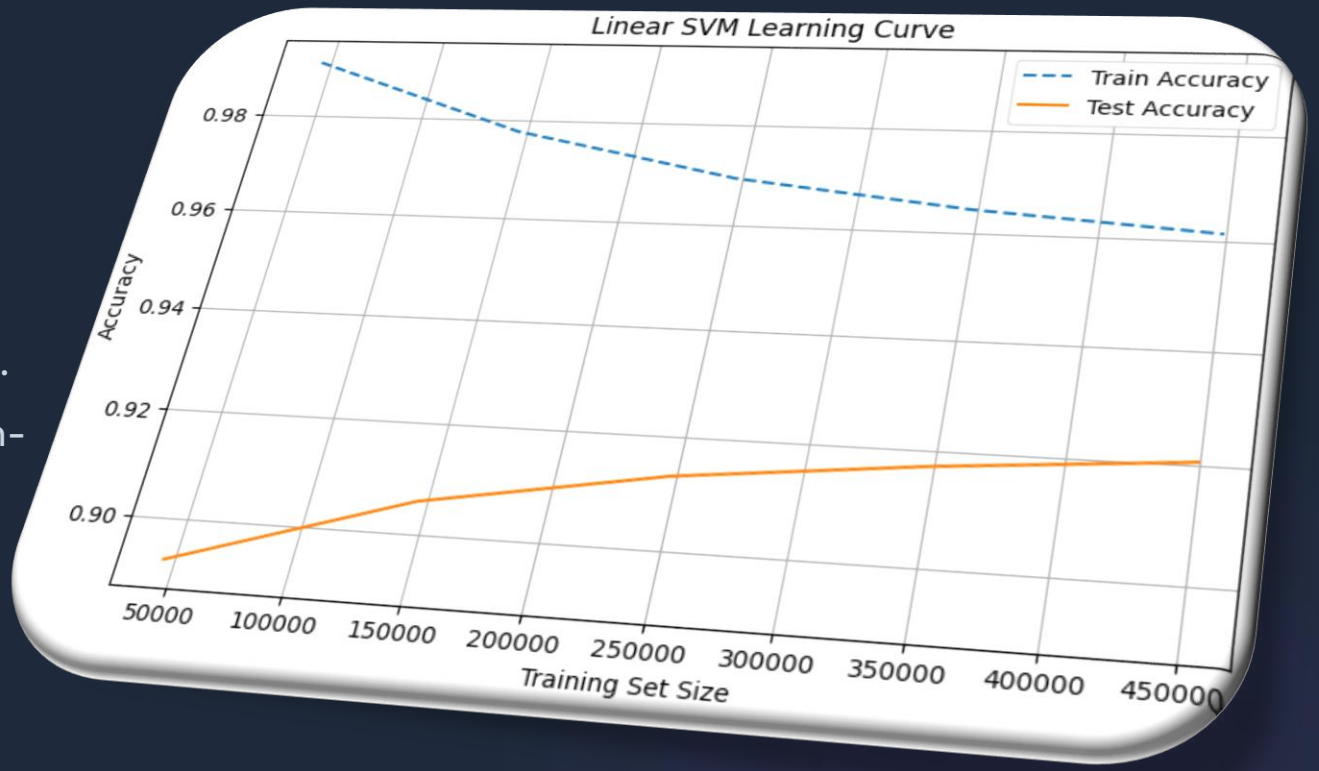
Class	Precision	Recall	F1-Score	Support
0 (Not Hate)	0.95	0.97	0.96	141,728
1 (Hate)	0.81	0.71	0.75	26,739
Accuracy	**0.93**			

Insight: The clear winner. Recall jumped to 0.71, detecting ~18,800 toxic comments compared to ~15,000 for others.

Learning Curve

Training Analysis

- Best generalization gap among all models.
- Robust against overfitting despite the large feature space.
- Maximizing the margin allowed it to better handle the high-dimensional sparse data.



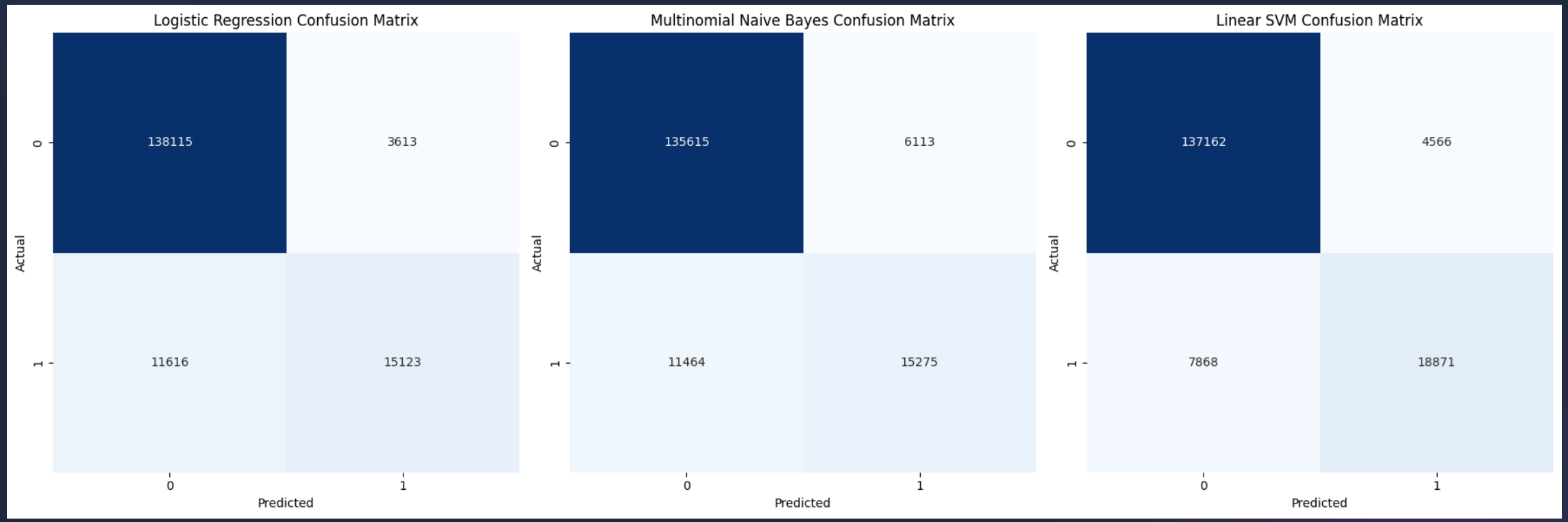
Confusion Matrix Analysis

False Negatives (The Risk)

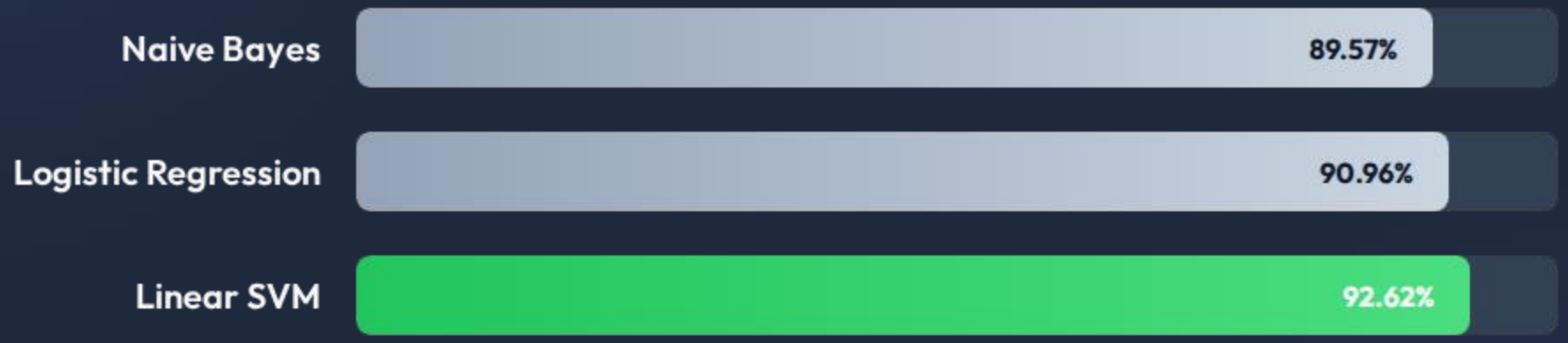
Logistic Regression: Missed 11,616 toxic comments.

Linear SVM: Reduced misses to 7,868.

SVM provides a safer environment by catching 3,600+ more toxic instances than the others.

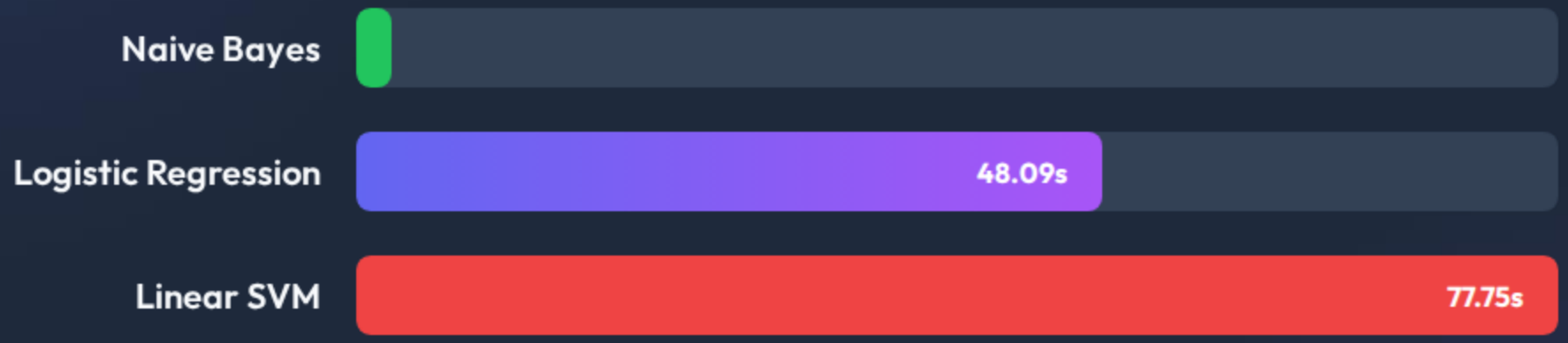


Comparative Analysis: Accuracy



While all models have high accuracy, SVM leads with 92.62%.

Computational Efficiency (Training Time)



Trade-off: SVM is ~130x slower than Naive Bayes, but 77 seconds is negligible for offline training given the performance boost.

Prediction on External Text

Input Text	Naive Bayes	Logistic Regression	Linear SVM
“We are worthless"	Hate	Hate	**Hate**
"I love this country"	Not Hate	Not Hate	**Not Hate**
"You are stupid idiot"	Hate	Hate	**Hate**
"Have a great day"	Not Hate	Not Hate	**Not Hate**

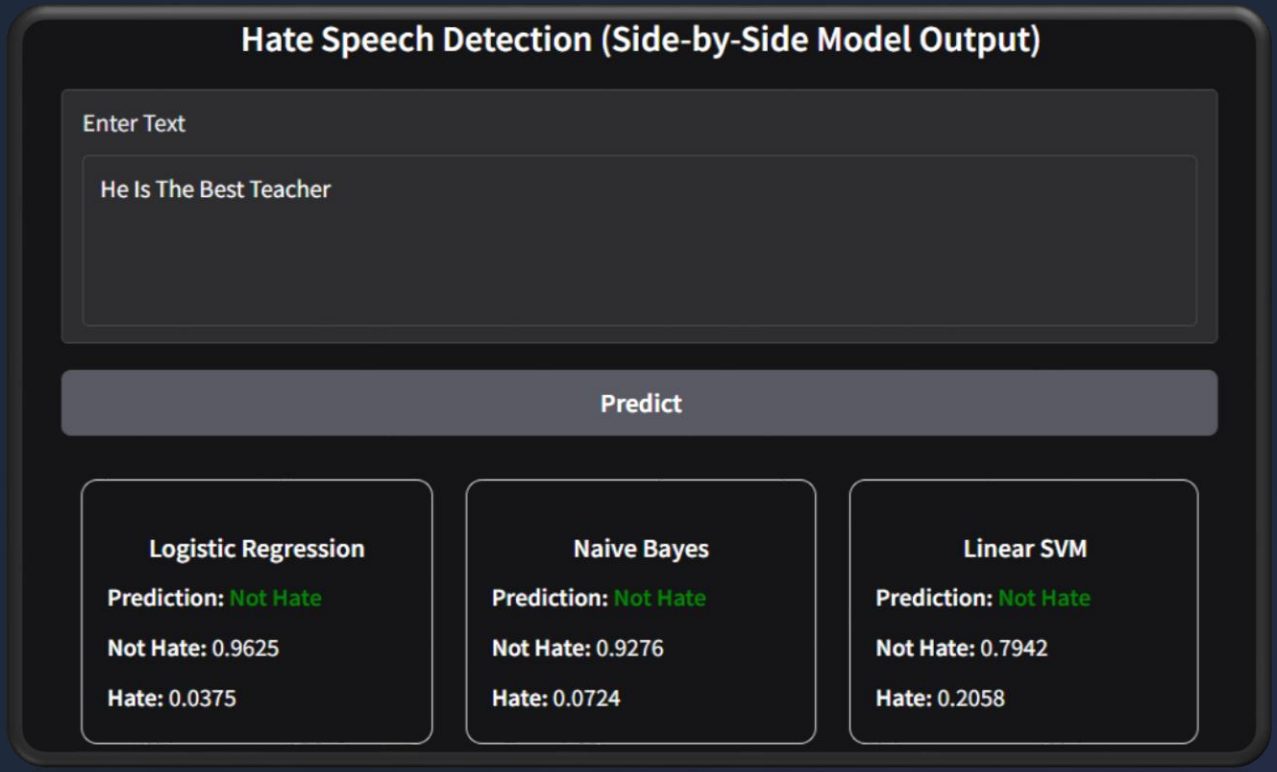
All models successfully distinguished between sentiment and didn't just flag "negative" words, proving semantic understanding.

Real-Time Deployment

Gradio Web Interface

We deployed the models using Gradio to simulate a real-world moderation dashboard.

- Allows side-by-side comparison.
- Visualizes confidence scores.
- Enables human-in-the-loop verification for edge cases.



Conclusion & Future Work

Conclusion

The Linear SVM proved to be the most robust model, offering the best balance between precision and recall (Accuracy: 92.62%).

We successfully deployed the model using a **Gradio web interface**, enabling real-time side-by-side comparison of model predictions.

Future Work

- **Deep Learning:** Implement BERT for better context understanding.
- **Data Augmentation:** Use SMOTE to fix class imbalance.
- **Multiclass:** Detect specific types (Racism vs. Sexism).

Thank you for your attention.